Practical No. 17: * Write a program to implement states of React Hooks

IX. Conclusion

Answer:
React Hooks, especially `useState()`, help manage dynamic data in functional components. It allows updating values and automatically re-rendering the UI. This makes code simple, reusable, and efficient without using class components.

X. Practical Related Questions

Q1. Write a React program to change the background color of a div using useState

Answer (program1.js)

```
import React, { useState } from "react";

function Program1() {
  const [color, setColor] = useState("lightblue");

  return (
    <div style={{ textAlign: "center" }}>
      <div
        style={{
          width: "200px",
          height: "200px",
          margin: "20px auto",
          backgroundColor: color,
        }}
      ></div>

      <button onClick={() => setColor("yellow")}>Change Color</button>
    </div>
  );
}

export default Program1;
```

Q2. Write a React program to manage user input using useState

Answer (program2.js)

```
import React, { useState } from "react";

function Program2() {
  const [name, setName] = useState("");

  return (
    <div style={{ textAlign: "center" }}>
      <h2>Enter Name:</h2>

      <input
        type="text"
        value={name}
        onChange={(e) => setName(e.target.value)}
      />

      <h3>Hello, {name}</h3>
    </div>
  );
}

export default Program2;
```

App.js (Run one program at a time)

```
import React from "react";
import Program1 from "./program1";
import Program2 from "./program2";

function App() {
  return (
    <div>
      {/* Run Program 1 */}
      {/* <Program1 /> */}

      {/* Run Program 2 */}
      <Program2 />
    </div>
  );
}

export default App;
```

XI. Exercise

Q1. What is the purpose of the useState() hook in React?

Answer:
The `useState()` hook is used to create and manage state in functional components. It allows storing values and updating them when needed.

Q2. Why do we use a setter function like setCount in React Hooks?

Answer:
The setter function (like `setCount`) is used to update the state value. When it is called, React re-renders the component and displays the updated data.